

PRONoy DAS

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PhD candidate in ECE at Purdue specializing in theoretical, computational, and experimental quantum photonics and quantum metrology. Seeking positions in areas of nanophotonics, quantum sensing, and computational electromagnetic modeling.

Education

Purdue University

Aug 2021 – Present

Ph.D. in Electrical & Computer Engineering (ECE), GPA: 3.89/4.0

West Lafayette, IN

- **Donnan Fellowship (2026)**: Purdue University - Recognition for exceptional performance in doctoral research.
- **APS Distinguished Student Award (2025)**: American Physical Society, Forum on International Physics
- **Purdue ECE Award for Excellence in Teaching (2024)**: Dept. of ECE, Purdue University
- **Published multiple first-authored articles on quantum photonics and quantum materials.**

Indian Institute of Science Education and Research (IISER) Kolkata

Aug 2016 – Jun 2021

Bachelor and Master in Physics, GPA: 8.82/10

West Bengal, India

Experience

Research Assistant | Dept. of ECE

Aug 2021 – Present

Birk Nanotechnology Center, Purdue University

West Lafayette, IN

- Created C++ and Python modules for Purdue-PicoMax to model optical responses and photonic dispersion in chiral semiconductors. Reduced modeling error to < 10% from 50% by using first-principles calculations in quantum photonics.
- Devised an ultrafast (GHz) modulation laser setup for twisted light using temperature-controlled Sagnac interferometry. Outcome: robust twisted-light generation for metrology and imaging.
- Discovered new fundamental quantum fluctuations in the orbital angular momentum of spatiotemporal optical vortices using low-photon Poissonian statistics + quantum optics.
- Generated shallow NV defects (30nm from surface) in diamond using ion implantation and vacuum anneal. Partnered to build a cryogenic quantum magnetometry setup using these single NV centers.

Graduate Course Instructor

Aug 2023 – May 2024

Dept. of Electrical Engineering, EE Fundamentals Lab I, Purdue University

West Lafayette, IN

- Led 60+ members with a team of 6 TAs, to simulate, prototype and analyze circuits for digital and analog logic projects.
- Recognized by the Purdue ECE Excellence in Teaching Award.

Undergraduate Researcher

Aug 2019 – Jul 2021

Dept. of Physics, IISER Kolkata

West Bengal, India

- Constructed a custom optical metrology setup with NV centers: developed confocal and ODMR microscopy, integrated lock-in detection, and produced image-analysis code in Python/Matlab for spectral fitting and denoising.
- Devised on-chip microwave antenna designs and performed EM simulation in 2-4 GHz in CST Studio Suite, and fabricated these antennas using UV photolithography for use in home-built cryogenic ESR setup.

Teaching Assistant

Aug 2020 – May 2021

Dept. of Physics, Quantum Information Processing & Classical Mechanics, IISER Kolkata

West Bengal, India

- Guided 30-40 students over each semester with their understanding of core course concepts.
- Assessed students' progress by designing and grading examinations and assignments.

Outreach and Leadership

Scientific Reviewer for Numerous High-Impact Journals

Aug 2024 – Present

- Reviewer for *Physical Review Applied* and *Physical Review Research* (APS Publishing), *New Journal of Physics* (IOP Publishing), *Proceedings of the Royal Society A*, and *SciPost Physics*

Research Implementation and Dissemination (multiple locations)

Aug 2021 – Present

- Published multiple articles in high-impact journals, accumulating over 10,000 downloads.
- Collaborated in writing multiple research grants, produced detailed technical reports for sponsors, and presentations for academic and non-academic audiences.

Team Lead, Multimedia and Web Technology, INQUIVESTA, IISER Kolkata

Aug 2017 – Mar 2018

- Led a team of 8 members towards front-end web development, sponsor meetings, graphic design and media coverage.

Skills

- **Programming & tools**: Python, C++ & Matlab, HTML & LaTeX, COMSOL and CST Studio.
- **Quantum Techniques & Optics**: ODMR & ESR spectroscopy, confocal microscopy and imaging, ultrafast (GHz) laser modulation, lumped circuit antenna design for qubit control, creating spin-defects in crystals.
- **Communication**: Strong verbal skills, collaborative teamwork, technical reporting and writing grants.

Hobbies: Film camera restoration, photography and development, IR photography, mechanical watch designing and assembly